

SMART MIRROR

An Interactive Reflective Interface for Daily Information Management

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INTRODUCTION

Traditional mirrors serve only the purpose of reflection, requiring users to rely on multiple devices such as smartphones, tablets, and smartwatches to access daily information. This is inconvenient, especially during busy routines. The Smart Mirror addresses this issue by combining a standard two-way reflective surface with a digital display behind the glass, centralizing essential information - time, weather, calendar events, and to-do lists - into a single, easily accessible interface. The system leverages IoT and Embedded Systems technologies to deliver a modern, energy-efficient, and user-friendly smart home device that improves daily productivity and convenience.

SUMMARY

- Centralizes daily information (time, weather, calendar, and to-do lists) into a single, hands-free interface.
- Integrates an HD digital display seamlessly behind a two-way reflective surface.
- Utilizes a PIR motion sensor to automatically toggle the display, ensuring energy efficiency.
- Enhances daily productivity by minimizing the need to check multiple mobile devices during busy routines.

RESULT

The final integrated output. It acts as a standard mirror while seamlessly allowing the bright digital information to shine through when activated.



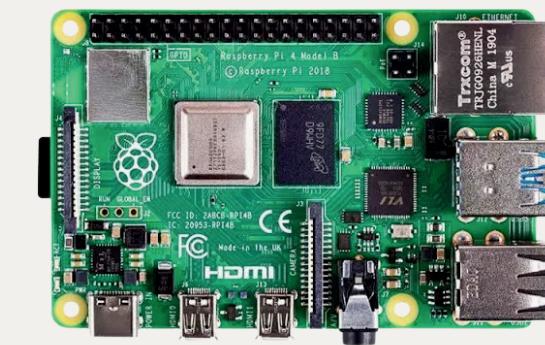
Two-Way Mirror Interface

Provides an interactive, high-contrast visual output. It allows users to directly navigate the digital dashboard via touch and receives data from the Raspberry Pi.



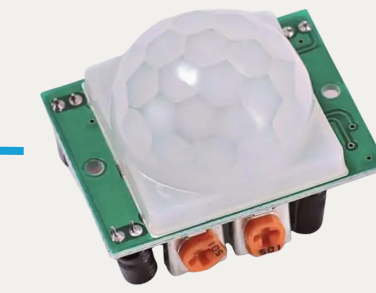
15.6" HD Touch Display

The core processing unit. It receives the sensor's signal, runs the MagicMirror² framework, and fetches real-time API data from the internet.



Raspberry Pi

Detects user presence and sends a signal to the Raspberry Pi to activate the display, ensuring energy efficiency.



PIR Motion Sensor

CONCLUSION

- Demonstrated the successful integration of embedded systems and IoT to create a functional and practical smart home device.
- Validated the system's energy efficiency, driven by the motion-sensor auto-wake/sleep mechanism, keeping operating costs minimal.
- Confirmed that the Raspberry Pi 5 platform provides the optimal balance of performance, flexibility, and cost for this project.
- Identified potential future enhancements, including multi-user profiles, voice assistant integration, and mobile app synchronization for expanded functionality.

METHODOLOGY

- Utilizing a Raspberry Pi 5 as the central processing unit to process sensor data and fetch real-time information via internet APIs.
- Employing the MagicMirror² framework on Raspberry Pi OS to manage and customize the widgets on the digital dashboard.
- Integrating a PIR motion sensor to detect user presence, automatically toggling the display to conserve energy.
- Combining a 15.6-inch HD monitor with two-way mirror glass to create a seamless, interactive reflective surface.

